

2	(a)	Simple harmonic motion is an oscillatory motion in which the acceleration is directly proportional to the displacement from an equilibrium position, and acceleration is always directed towards that point. (or acceleration is in the opposite direction to the displacement).		B1 B2
	(b)	(i)	Since g and r are both constants, the equation is of the form $y = -kx$, hence the <u>acceleration of the ball is directly proportional to its displacement</u>. Also, the negative sign indicates that the ball's acceleration is in the opposite direction to its displacement. Hence ball undergoes SHM.	B1 B2
		(ii)	Ball is displaced by a small angle θ , or The radius of curvature is large, ie small angle approximation is applied in the derivation.	A1
		(iii)	$\omega^2 = \frac{g}{r} \quad [\text{M1 with correct substitution}]$ $\omega = \sqrt{\frac{9.81}{0.28}} = 5.919 \text{ rad s}^{-1}$ $\frac{2\pi}{T} = 5.919$ $T = 1.0615 \text{ s} \quad [\text{M1}]$ The time interval to pass P and return to it is $\frac{T}{2}$ $\therefore \tau = \frac{1.0615}{2} = 0.531 \text{ s} \quad [\text{A1}]$	M1 M1 A1
		(iv)	1. $s = r\theta$ $= (0.28)\left(\frac{1}{360} \times 2\pi\right)$ $= 4.8869 \times 10^{-3} \text{ m} \quad [\text{M1}]$ Assuming $x_0 \approx s = 4.8869 \times 10^{-3} \text{ m}$	M1

			2.	$\text{Max PE} = \frac{1}{2} m \omega^2 x_0^2$ $= \frac{1}{2} (0.005)(35.03)(4.8869 \times 10^{-3})^2$ $= 2.09 \times 10^{-6} \text{ J}$ Alternatively, $\text{Max PE} = mgh$ $= (0.005)(9.81)(0.28 - 0.28 \cos 1^\circ)$ $= 2.09 \times 10^{-6} \text{ J}$	C1 A1
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